Dynamic Pricing and Demand Forecasting for Sports Facility Aggregators: Methods and Materials

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Abstract—This document presents the Methods and Materials section (Assignment 2) of a study on dynamic pricing and demand forecasting for sports facility aggregators. It specifies the quantitative experimental design, the data source and its access and ethics conditions, preprocessing, a strategy for identifying price elasticity when historical prices are endogenous, a leakage-free temporal validation and tuning protocol, probabilistic evaluation metrics suitable for a binary booking target, a consistent counterfactual revenue metric, pre-specified statistical success criteria, and threats to validity.

I. METHODS AND MATERIALS

To address the research gap identified in the literature review [9], [17]—the lack of an integrated short-horizon demand forecasting and dynamic pricing framework for bookable, capacity-constrained sports facilities—this study employs a quantitative, data-driven counterfactual simulation strategy [1], [2]. The operational objective is to jointly model price-sensitive booking probability conditional on exogenous environmental features [6] and to optimize slot-level expected revenue within the price boundaries set by facility owners [8].

A. Research Design and Methodological Strategy

The research strategy is a quantitative design structured as a two-stage sequential pipeline [4]. A purely survey-based inquiry is insufficient because bookings depend on temporal dependencies, sudden exogenous shocks (rain, heatwaves) [6], and price sensitivity that can only be captured through formal modeling and counterfactual evaluation [3]. The architecture has two decoupled modules:

1) **Stage 1: short-horizon booking-probability forecasting.** For each venue v and slot t, the module estimates the calibrated probability $D_0(x_t) = P(y_t = 1 \mid p_t = p_{ref,t}, x_t)$ that the slot is booked at its reference (historical) price, where x_t contains lead time, hour, weekday, weather, calendar flags and venue attributes [12], [14].
2) **Stage 2: constrained dynamic pricing.** A price-response layer (Section C) converts D_0 into $D(p, x_t)$ for alternative prices, and the optimizer outputs the recommended price p_t^* subject to owner constraints [8], [9].

The modular design preserves interpretability and stability, and allows facility managers to inspect forecasts before prices are applied. End-to-end reinforcement learning is avoided because it tends to produce erratic pricing in sparse-data settings [4]. Stage 2 maximizes expected slot profit:

$$\Pi_t(p) = (p - c_t - \varphi_t\, q_t(p, x_t)) \cdot D(p, x_t) \quad (1)$$

$$p_t^* = \arg\max_{p \in P(t)} \Pi_t(p) \quad (2)$$

where c_t is the variable cost per booking, q_t the cancellation probability given a booking, φ_t the expected refund or penalty cost per cancellation [5], and $P_t = \{p$ on a price grid : $p_{min} \le p \le p_{max}$, p within the observed price support of the segment$\}$. The bounds p_{min}, p_{max} are set by facility owners [8]; restricting to the observed support prevents extrapolating the demand model outside the data. Because D is a black-box function of p, (2) is solved exactly by enumerating the discretized grid (step of 1% of p_{ref}).

B. Data Sources, Access and Ethics, and Preprocessing

1) Primary transactional dataset. The data come from [AGGREGATOR NAME], a sports facility aggregator operating [N] venues in [city/region], and are provided under a data-sharing agreement [agreement reference and date]. The dataset contains approximately 150,000 booking records for [start date]–[end date] across indoor futsal arenas, outdoor football pitches, tennis courts and swimming lanes. Booking records alone contain only realized bookings, so they are joined with the slot-inventory table (all published slots, including those that stayed empty; [N_slots] rows) to obtain the binary label y_t and the unbooked negatives. Each slot record has venue and slot identifiers, start/end time, booking and cancellation timestamps (for lead time), list price, final paid price, discounts and status [7].

Privacy and ethics. Customer identifiers are removed or pseudonymized by the aggregator before transfer; the analysis uses no personal data, in line with the Law of the Republic of Kazakhstan on Personal Data and Their Protection [29]. Venue identities are anonymized and results are reported in aggregate to protect commercial confidentiality. Ethics review status: [approval/exemption reference of the university ethics committee].

2) Weather and calendar covariates. Weather is obtained from the Open-Meteo APIs [18], which are free for non-commercial research use and licensed CC BY 4.0 (attribution is given). Realized hourly weather comes from the ERA5 reanalysis [19]; weather *forecasts available at decision time* come from Open-Meteo's historical forecast archive (available from 2021), matched to venue coordinates. Variables: air temperature (°C), relative humidity (%), wind speed (m/s) and precipitation (mm/h). To avoid look-ahead leakage, model inputs at a given lead time use the latest forecast issued before the decision time; realized (ERA5) weather is used only in a robustness check that provides an upper bound on accuracy. Calendar attributes include national holiday flags and academic examination periods from university calendars [15].

3) Preprocessing and feature engineering [11].

a) *Cleaning and imputation.* Duplicate records and invalid prices are removed. Weather gaps are filled by linear interpolation (gaps ≤ 3 h) and forward-fill (≤ 2 h). All imputation statistics are computed on training data only.
b) *Cyclical encoding.* Hour of day, day of week and month are encoded with sine and cosine terms (period T) to preserve periodic boundaries:

$$x_{sin} = \sin(2\pi t/T),\ x_{cos} = \cos(2\pi t/T) \quad (3)$$

c) *Lead-time bins.* Booking lead time is categorized into decision windows (< 3 h, 3–24 h, 1–7 days, > 7 days) to model changing price sensitivity as slot expiration

approaches [9].

d) *Relative price.* Prices enter as $r_t = p_t / p_{\text{ref},t}$, where $p_{\text{ref},t}$ is the venue's list price for that slot type, so that price effects are comparable across venue types.

C. Demand Model and Identification of Price Elasticity

Identification problem. Historical prices were set by owners in anticipation of demand (higher prices in expected peaks). A regression of bookings on observed prices therefore confounds price effects with demand conditions and biases elasticity toward zero or even to the wrong sign. Estimated elasticity is not taken from historical correlations alone; the following strategy is used.

1) **Orthogonalized estimation.** A partially linear model $y_t = \theta(z_t) \ln r_t + g(x_t) + \varepsilon_t$, with $\ln r_t = m(x_t) + v_t$, is estimated by double/debiased machine learning [20]. The nuisance functions g and m are fitted with LightGBM [27] under time-blocked cross-fitting, and θ is identified from the residual price variation v_t. The group variable z_t covers venue type, lead-time bin and peak/off-peak status. The share of price variance not explained by x_t is reported; if it is negligible for a segment, that segment is excluded from dynamic pricing.
2) **Quasi-exogenous episodes.** Together with the aggregator, price changes whose timing is plausibly unrelated to short-run slot demand (seasonal price-list updates, platform-wide promotions) are catalogued and analyzed with event-study / difference-in-differences designs against unchanged venues, with a pre-trend test.
3) **Instruments.** Cost shifters (e.g., utility tariff changes) are used as instruments only if the first stage is strong ($F > 10$) and exclusion is plausible; otherwise they are not used.
4) **Price pilot (if permitted).** A small randomized price perturbation on a subset of off-peak slots, with owner consent, is used to benchmark the estimated elasticities.
5) **Sensitivity.** Elasticities are scaled by factors 0.5–1.5, and the elasticity bias that would eliminate the revenue gain (break-even bias) is reported.

Price-response layer. The identified elasticity $\varepsilon(z) = \theta(z)/\bar{D}(z)$ (semi-elasticity divided by mean booking probability of the group; valid for moderate price changes within the observed support) is combined with the Stage 1 forecast:

$$D(p, x_t) = \min\{1, D_0(x_t) \cdot (p/p_{\text{ref},t})^{\varepsilon(z)}\} \tag{4}$$

This separates *prediction* (Stage 1, trained on observed data) from *causal price response* (identified above). A price-feature version of the forecasters with a monotonic non-increasing constraint on price is also trained, and its implied elasticity is compared with θ as a consistency check.

D. Validation Protocol, Tuning, and Experimental Setup

1) Temporal split. All splits are chronological; no shuffling is used. The final ≈ 20% of the timeline is a held-out test set that is used once, after all design decisions. The earlier ≈ 80% is used for expanding-window rolling-origin cross-validation with $K = 4$ folds [25], with a 7-day gap between training and validation windows to prevent leakage through adjacent slots and lagged features. Because seasonality in Astana is strong and the data span about two years, results are reported per season, and a leave-venue-out evaluation assesses generalization to unseen venues.

2) Leakage control. Scalers, encoders and imputers are fitted on training folds only; only information available at the decision time is used as input (including forecast, not realized, weather); lag features use past data only.

3) Models compared. Pricing baseline 0: historical fixed pricing set by facility owners. *Stage 1 candidates* (all receive the same features): a logistic regression (reference GLM); SARIMAX on venue-level hourly occupancy with weather regressors [11]; LightGBM [6], [27]; and LSTM / Temporal Fusion Transformer [12], [13], [28]. The Stage 1 model for Stage 2 is selected on validation log-loss.

4) Hyperparameter tuning. Each model gets the same budget (100 Bayesian-optimization trials, Optuna [30]) on the rolling-origin folds, minimizing log-loss, with early stopping for GBDT and neural models. SARIMAX orders are chosen by AIC. Every neural and GBDT result is the mean ± s.d. over 5 random seeds.

5) Software and hardware. Python 3.11 with scikit-learn, LightGBM, statsmodels, PyTorch and DoubleML/EconML [16], on Ubuntu 22.04 LTS with an NVIDIA RTX 4090 GPU (24 GB) and 64 GB RAM. Seeds, configurations and code are versioned for reproducibility.

E. Evaluation Metrics and Success Criteria

1) Forecasting metrics. The target $y_t \in \{0, 1\}$ is binary and the prediction $\hat{y}_t$ is a probability. MAPE is undefined when $y_t = 0$ (division by zero) and is discouraged for such data [23]; it is therefore not used at slot level. Proper scoring rules [21], [22] are used instead:

$$\text{Brier} = (1/N) \sum_{t=1}^{N} (y_t - \hat{y}_t)^2 \tag{5}$$

$$\text{LogLoss} = -(1/N) \sum_t [y_t \ln \hat{y}_t + (1 - y_t) \ln(1 - \hat{y}_t)] \tag{6}$$

$$\text{ECE} = \sum_{b=1}^{B} (n_b/N) \cdot |\text{acc}(b) - \text{conf}(b)| \tag{7}$$

In addition: ROC-AUC (ranking quality), the calibration slope and intercept with reliability diagrams [24], and the Brier skill score relative to a climatological baseline (mean booking rate per venue × hour × weekday). For comparison with SARIMAX, which models aggregated series, predictions are aggregated to venue-day occupancy $Y_{v,d}$ and evaluated with MAE and the weighted absolute percentage error, which is defined whenever total occupancy is positive:

$$\text{WAPE} = \sum_{v,d} |Y_{v,d} - \hat{Y}_{v,d}| \,/\, \sum_{v,d} Y_{v,d} \tag{8}$$

2) Business and operational metrics. Revenue is evaluated by counterfactual offline simulation on the test period. Observed revenue ($p_0 y$) is *not* compared with expected revenue under the model, because they are different quantities. Both policies are evaluated with the same demand model, with $p_{0,t}$ the historical price and $R(\pi) = \sum_t \pi_t D(\pi_t, x_t)$:

$$\Delta R = [R(p^*) - R(p_0)] \,/\, R(p_0) \times 100\% \tag{9}$$

The demand model is calibrated on the training period so that $R(p_0)$ matches observed revenue there; the ratio of $R(p_0)$ to observed revenue on the test period is reported as a diagnostic. Uncertainty is quantified by venue-week block bootstrap (95% CIs), propagating elasticity uncertainty by Monte Carlo draws from the sampling distribution of $\hat{\theta}$ (1,000 draws). Other metrics:

- **Capacity utilization:** $U = \sum_t D(p_t^*, x_t) \,/\, N_{\text{slots}} \times 100\%$, compared with the historical booking rate.
- **Off-peak occupancy:** booking rate on weekdays 10:00–16:00, a window defined in advance and verified on training data to be the lowest-occupancy period.
- **Cancellation rate:** predicted by a classifier $q_t(p, x_t)$ and compared with the historical rate.

3) Pre-specified success criteria. Criteria are fixed before test-set evaluation and rely on statistical and practical

significance rather than arbitrary absolute thresholds, with Holm correction for multiple comparisons:

1) **Forecasting.** The selected Stage 1 model has lower Brier score and log-loss than every baseline on the test set (paired block bootstrap / Diebold–Mariano [26], adjusted $p < 0.05$), a positive Brier skill score, and a calibration slope whose 95% CI contains 1.
2) **Revenue.** The lower bound of the 95% CI of ΔR exceeds ΔR_{min}, the smallest gain that covers the implementation cost, agreed with the aggregator [ΔR_min] before the test run. If no such value exists, $\Delta R_{min} = 0$ and ΔR is reported as an effect size with its CI and the minimum detectable effect computed from training-period variance (80% power).
3) **Off-peak and cancellations.** Off-peak utilization increases (CI lower bound > 0) while cancellations are non-inferior: the upper 95% bound of (new − historical) cancellation rate is below a margin equal to half the standard deviation of the historical weekly cancellation rate (i.e., within normal fluctuation).

F. Limitations and Threats to Validity

- **Regionality and climate.** Data reflect Astana (−30 °C to +35 °C), giving a strong weather signal [6]; findings may require recalibration for temperate zones. Using one aggregator and about two years of data limits generalization across seasons and platforms.
- **Counterfactual assumptions.** Offline revenue relies on estimated elasticities [10]; residual confounding may remain despite the identification strategy of Section C, and real customers may react differently to novel dynamic pricing (learning, strategic waiting). Competitor prices are unobserved. Sensitivity analysis and the optional price pilot mitigate but do not remove this risk.
- **Weather data.** Reanalysis and forecast grids (about 0.1°–0.25°) are coarser than venue locations; forecast errors reduce achievable accuracy relative to realized weather.
- **Owner governance and churn.** Aggressive price changes can cause dissatisfaction among owners or users [8]; this is mitigated by owner-controlled price floors and ceilings and by restricting prices to the observed support.
- **Data access.** Commercial confidentiality may restrict data sharing; replication is supported through code and aggregated outputs.

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